Echo Game Design Document

Alpha Progress Check | Team Missing Piece

Team Information

Name	Roles	Email	GitHub ID
Samarth Rajendra	Team Captain; gameplay and AI programmer	sr47937@usc.edu	samarthraj7
Jeevika Kiran	Product Manager; scoring and analytics programmer	jeevikak@usc.edu	JeevikaK
Rhesha Susil Vinod	Stenographer; UI and controls programmer	rvinod@usc.edu	RheshaVinod
Sanath Nagendra Bhargav	Level designer; switch and arena programmer	sanathna@usc.edu	Sanath-07
Prathiksha Ravibabu Nijamkari	Ghost systems and visual feedback programmer	ravibabu@usc.edu	Prathiksha3008

We all contribute to game development and coding in addition to our assigned coordination roles.

Project Links

Deliverable	Location
Unity Project GitHub Repository	GitHub repository
Playable Alpha Progress Check Build	Play Echo
Alpha Gameplay Video	EchoDemoVideo.mp4 (2 minutes 21 seconds; included in submission folder)

Game Introduction

Logline

Echo is a competitive grid-based treasure race in which players commit to full-board slides while their previous moves return as looping ghosts that can block routes, flip walls, and collect gems.

Player Goals and Win Conditions

- Collect 5 gems before the opponent. Sliding through a gem collects it, and a player's ghost can collect gems for its owner.
- If one player is boxed in, that player may spend 1 gem to respawn near home or skip the turn.
- If both players are boxed in, the player with more gems wins; equal scores produce a draw.

Game Description

Echo is a turn-based, two-player grid strategy game played on a 9 by 9 arena. On each turn, the active player chooses a direction and slides until a wall, player, or ghost stops the token. Moving through the green gem scores a point. Yellow switches toggle every purple wall between a raised, blocking state and a lowered, passable state, so each slide can change the routes available to both players.

The game's twist is an owned looping ghost. After every five slides, a ghost appears near the start of that sequence and repeatedly replays those five directions, taking one step after each move by its owner. Ghosts occupy space, trigger switches, and collect gems for their owner. Because walls and blockers change over time, the same recorded direction can produce a different result on a later loop. The current Alpha supports one ghost per player, three bot difficulties, local hot-seat play, undo, restart, and boxed-in recovery.

Controls

Input	Action
Arrow keys, WASD, swipe, or on-screen arrows	Tilt the board; the active token slides until stopped
Z or Undo	Undo the last move and the bot response
R or Restart	Start a new match
1, 2, 3, 4 or mode buttons	Easy bot, Medium bot, Hard bot, or local 2 Players
H	Cycle the mode and start a new match
G	When boxed in, spend 1 gem and respawn near home
N	When boxed in, skip the turn

Detailed Design

Game Mechanics

Mechanic	Implemented Alpha Behavior
Slide until stopped	A directional input commits the active token to a straight path until a wall, player, or ghost blocks it.
Global switch walls	Crossing a yellow switch toggles every purple wall between raised and lowered states.
Gem scoring	A player token or owned ghost scores when it passes through the green gem. The first player to 5 wins.
Echo recording	Every group of five player slides becomes a repeating ghost sequence.
Ghost interaction	Ghosts replay one recorded direction after each owner move, block all pieces, trigger switches, and score for their owner.
Boxed-in recovery	A blocked player may spend a gem to respawn or skip. If both are blocked, scores resolve the match.

Game Mechanic Research

We each researched a different mechanic relevant to our selected prototype. The entries below use the four-question format required for the Alpha Progress Check.

Prathiksha Ravibabu Nijamkari

Game Title	Polarity Switch
Mechanic or System	Magnetic attraction and repulsion
Interest	Polarity turns simple block movement into a positioning puzzle in which direction and stopping points matter.
Interaction with Other Mechanics	Attraction and repulsion interact with walls, obstacles, and objectives, requiring the player to plan where a block will stop.
Innovation	The mechanic can become competitive when two magnets manipulate shared blocks toward opposing goals and switches change both players' polarities.

Jeevika Kiran

Game Title	Into the Breach
Mechanic or System	Telegraphed actions and deterministic grid interactions
Interest	Revealing outcomes before commitment creates strategic depth while keeping the rules readable and learnable.
Interaction with Other Mechanics	Telegraphed actions combine with grid movement, collision, displacement, objectives, hazards, and turn order, allowing one move to affect several systems.
Innovation	Echo applies readable prediction to a competitive board: recorded ghost directions are visible, but switches and blockers can change where those directions lead.

Samarth Rajendra

Game Title	Super Time Force and Ricochet Robots
Mechanic or System	Past-action copies and slide-until-stop movement
Interest	Past actions become active pieces, while committed sliding makes each direction a strategic choice rather than a distance selection.
Interaction with Other Mechanics	Recorded actions interact with later blockers and board geometry. A route that previously scored may later trigger a switch, stop early, or miss.
Innovation	Echo gives each player an owned ghost that loops five recorded directions, blocks both players, flips walls, and collects gems for its owner.

Sanath Nagendra Bhargav

Game Title	Pokemon Ice Path and Switchback
Mechanic or System	Committed sliding and a shared switch-wall state
Interest	Slide-until-stop movement is easy to learn, while global wall changes require players to plan for the board state they will reach.
Interaction with Other Mechanics	A switch can open a lane for the active player while closing it for the opponent or for a ghost that will replay later.
Innovation	Echo places two competitors, a gem, and looping ghosts in the same switch-controlled arena so every route can serve several future actors.

Rhesha Susil Vinod

Game Title	Pac-Man
Mechanic or System	Moving ghosts that occupy space and reshape routes
Interest	An autonomous blocker creates tension without adding another player command and turns a static maze into a changing spatial puzzle.
Interaction with Other Mechanics	Ghost occupancy interacts with maze topology and collectibles by closing safe corridors and forcing rerouting.
Innovation	Echo replaces AI pursuit with player-authored recordings, making the blocker a consequence of the opponent's past decisions or the player's own plan.

Chosen Concept

We prototyped both Magnet Grid and Echo. We selected Echo for the Alpha because slide movement, shared switches, gem scoring, and visible ghost replay produced a clearer core loop. Our innovative twist is that a player's plan returns and acts again on a board that may have changed.

Twist and Mechanics Matrix

Mechanic	Description	Interaction with Twist	Affected Genre Elements	Supports
Slide until stopped	A direction commits the token until blocked.	The five chosen directions become the next ghost's loop.	Movement, positioning, turn planning	Readable commitment
Owned looping ghost	Replays one of five recorded directions after each owner move.	The player's past plan becomes a new board actor.	Opponent blocking, timing, prediction	Core twist
Global switch walls	Any yellow switch toggles all purple walls.	The board may change before a ghost repeats the same input.	Maze topology, path access	Emergent outcomes
Gem race	Tokens and owned ghosts collect the shared gem; first to 5 wins.	A replay can score without the player's current token reaching the gem.	Competition, goals, risk	Match pressure

Mechanic Interaction Matrix

Subject	Slide Movement	Looping Ghost	Switch Walls	Gem Scoring
Slide movement	-	Records the five replay directions	Players can activate switches while sliding	Sliding through the gem scores
Looping ghost	The ghost's path uses the same stop rules	-	Ghosts can activate switches	Ghosts collect for their owner
Switch walls	Raised walls stop a slide; lowered walls do not	Changed walls alter replay outcomes	-	Walls gate routes to the gem
Gem scoring	The chosen path determines collection	A replay can score later	Switches open or close scoring routes	-

Sketches and Diagrams

Gameplay Loop

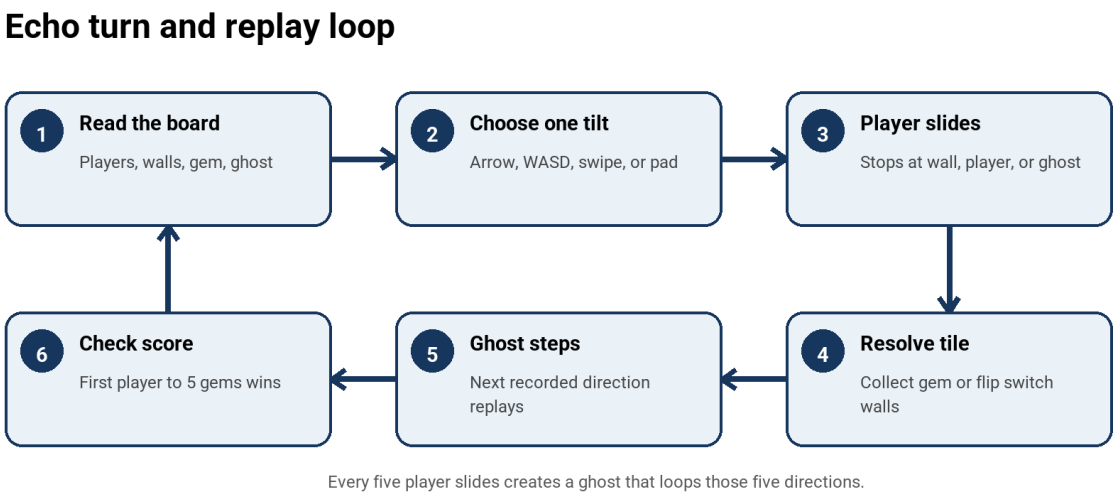


Figure 1. A player chooses one tilt, resolves the slide, then advances the owned ghost before checking the score. Every five player slides refreshes the ghost's repeating sequence.

Mechanics Diagram

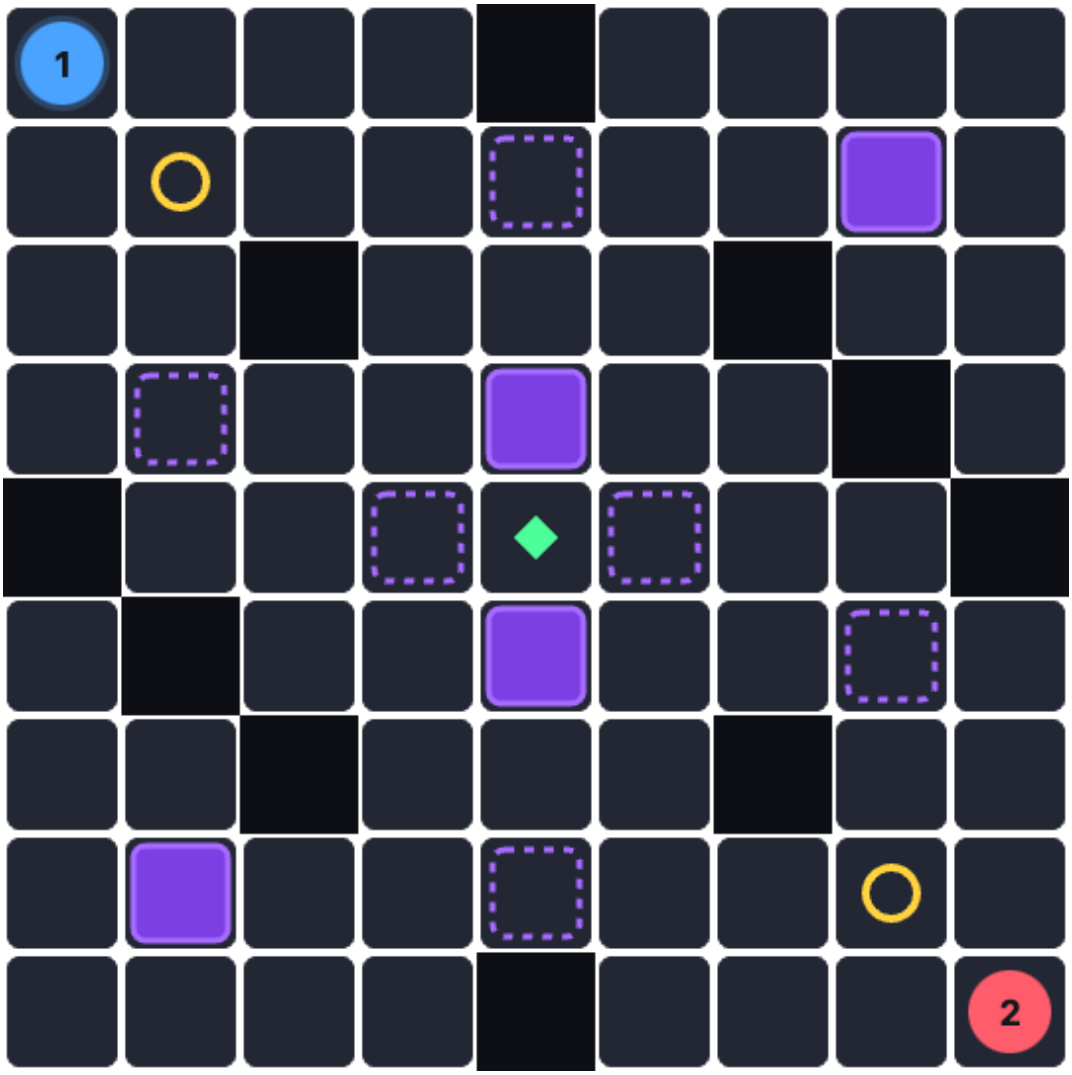


Figure 2. Implemented 9 by 9 arena. Blue starts in the upper-left, Red starts in the lower-right, yellow circles are switches, solid purple tiles are raised walls, dotted purple outlines are lowered walls, and the green diamond is the shared gem.

Alpha Status

Implemented Build

- One 9 by 9 arena with slide-until-stop movement, shared switches, raised and lowered walls, two player tokens, one shared gem, and move trails.
- A ghost after every five player moves, with one active ghost per player. Ghosts block pieces, trigger switches, and collect gems.
- Boxed-in recovery through spending a gem or skipping, plus score resolution if both players are stuck.
- Undo, restart, Easy, Medium, and Hard bots, and local two-player hot-seat mode.
- Unity 6 WebGL deployment on GitHub Pages.

Modes

Easy, Medium, and Hard use bots that search approximately 1, 2, and 4 moves ahead. Two Players is local hot-seat play on one keyboard. Online multiplayer is not part of the Alpha build.

Analytics Plan

We plan to add the following events to the WebGL build. Each event will include the selected mode and a match identifier. These events are planned and are not yet implemented in our Alpha.

Event	Trigger	What It Reveals
match_start	New game	Modes selected by players
tilt	Successful slide	Match length and move volume
gem_collect	Token or ghost scores	Whether ghosts contribute to scoring
ghost_spawn	Ghost appears	When players first reach the twist
boxed_in	No legal slide	Frequency of blocked states
unstick_spend or unstick_skip	Boxed-in choice	Whether spending a gem feels worthwhile
undo	Undo selected	Where confusion or mistakes occur
match_end	Win, draw, or restart	Winner, scores, turn count, and mode

Playtest Plan

Run 8 to 10 minute sessions against the Medium bot, followed by one local two-player match when time permits. Do not explain the ghost unless the tester asks. Record whether the HUD communicates the approaching ghost, whether players intentionally use a ghost, how boxed-in choices are interpreted, where players become lost, and whether they request more ghosts, gems, or space after one match.

Weekly Prototype Descriptions

Week 8

We established our selected concept and its rules: committed slide movement, a shared gem race, globally toggled walls, and a five-move ghost recording. We divided responsibilities across gameplay rules, arena layout, scoring, UI, and ghost feedback. Figure 1 documents the resulting gameplay loop and the interactions we intended to test.

Week 9

We assembled the playable Unity Alpha, added one ghost per player, completed gem scoring and boxed-in recovery, connected the HUD and controls, added three bot difficulties and local two-player play, and published our WebGL build to GitHub Pages. Our gameplay video demonstrates this integrated state.

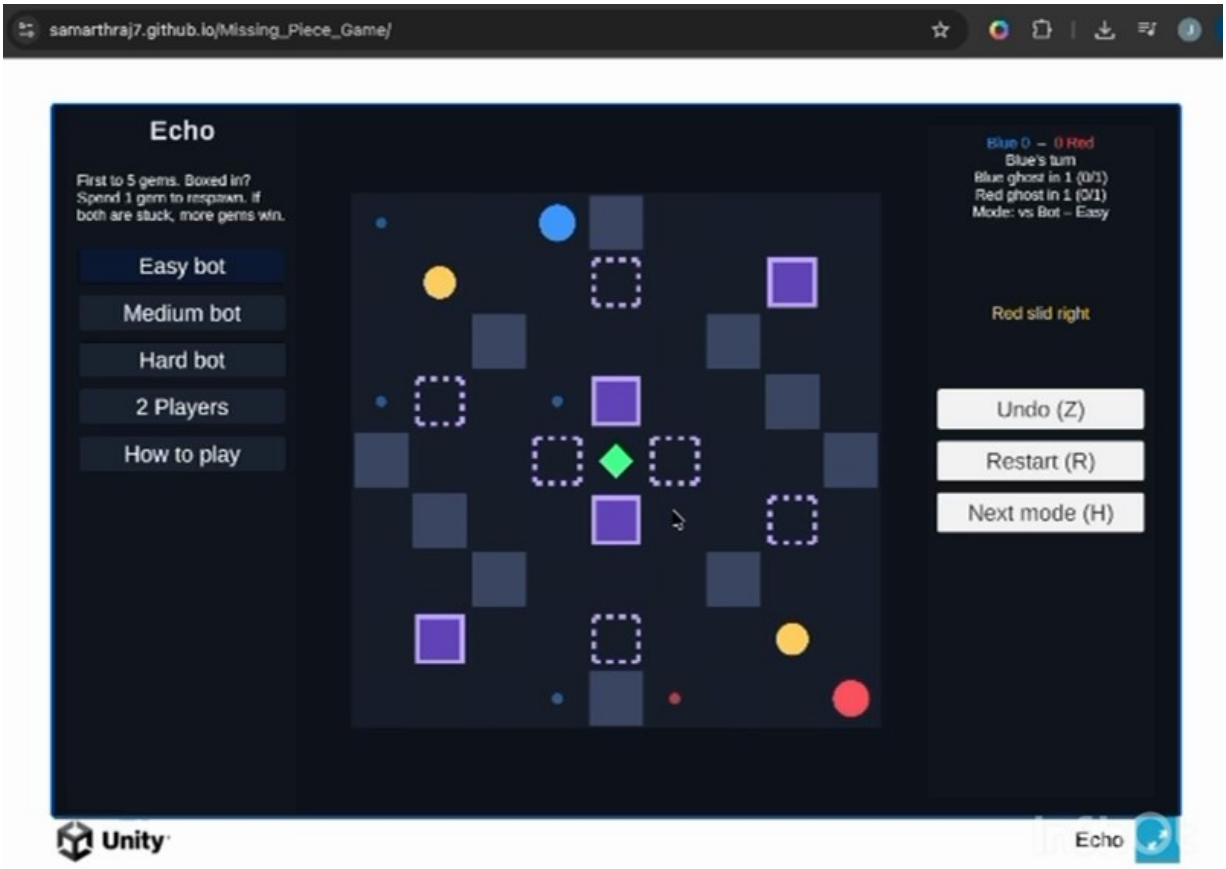


Figure 3. Week 9 Alpha build running from GitHub Pages. The full 2 minute 21 second recording is included as EchoDemoVideo.mp4.

Weekly Progress

Our public repository currently contains one combined implementation commit. The table records our documented work split and links the available commit without claiming separate individual commit histories.

Week	Team Member	Individual Work	GitHub Commit
Week 8	Samarth Rajendra	Defined core rules, Unity loop, bot approach, and WebGL plan.	No separate Week 8 commit is visible in the current public history.
Week 8	Sanath Nagendra Bhargav	Designed the arena structure and switch-wall layout.	No separate Week 8 commit is visible in the current public history.
Week 8	Jeevika Kiran	Defined gem spawning, scoring behavior, and analytics ownership.	No separate Week 8 commit is visible in the current public history.
Week 8	Rhesha Susil Vinod	Planned HUD, input affordances, and ghost countdown communication.	No separate Week 8 commit is visible in the current public history.
Week 8	Prathiksha Ravibabu Nijamkari	Designed ghost tokens, trails, and spawn feedback.	No separate Week 8 commit is visible in the current public history.
Week 9	Samarth Rajendra	Integrated rules, bot modes, the Unity gameplay loop, and the WebGL deployment.	Combined commit 9e1eb2f
Week 9	Sanath Nagendra Bhargav	Implemented the Alpha arena and switch-wall LevelData.	Combined commit 9e1eb2f
Week 9	Jeevika Kiran	Implemented gem spawning and first-to-five scoring.	Combined commit 9e1eb2f
Week 9	Rhesha Susil Vinod	Implemented the HUD, on-screen controls, undo, and restart wiring.	Combined commit 9e1eb2f
Week 9	Prathiksha Ravibabu Nijamkari	Implemented ghost tokens, trails, and spawn visuals.	Combined commit 9e1eb2f

External References

We used the following external tools and resources. We reviewed and edited all generated code and text. No generated art or audio ships in our build.

Name	Link	Usage
Cursor	https://www.cursor.com/	Assisted with drafting portions of the Unity C sharp rules, bot, HUD, and earlier GDD text. We reviewed and revised the output.
OpenAI Codex	https://openai.com/codex/	Assisted with reviewing the assignment, restructuring the GDD, editing prose, and formatting the final Word document. We reviewed the result.
Liberation Fonts	https://github.com/liberationfonts/liberation-fonts	Liberation Sans is used in the game under the SIL Open Font License.
Unity built-in primitives	https://docs.unity3d.com/6000.0/Documentation/Manual/PrimitiveObjects.html	Built-in Square and Circle sprites are used for board elements; no external art assets are included.